

ORF 245: Fundamentals of Statistics – Spring 2025

Precept 9

1. Linear Regression and Least Squares

(a) Statistical model

$$y = \beta_1 x + \beta_0 + \varepsilon$$

- Observe: x **explanatory variable**; y **response variable**.
- Not observed: ε **noise** (what the explanatory variable can not explain).
- Model parameters: β_1 and β_0 .
- Target: to estimate β_1 and β_0 using i.i.d. observations $(x_1, y_1), \dots, (x_n, y_n)$ sampled from the above model.
 - One critical assumption: **exogeneity** $\mathbb{E}[\varepsilon|x] = 0$.

(b) Linear Least Squares

- Optimization problem $(\hat{\beta}_1, \hat{\beta}_0) = \operatorname{argmin}_{\beta_1, \beta_0} \sum_{i=1}^n (y_i - \beta_1 x_i - \beta_0)^2$ (Training).
 - A geometric interpretation: find a line such that the total sum of the square of the difference between fitted value $\hat{y} = \hat{\beta}_1 x + \hat{\beta}_0$ and observed value y are smallest.
 - Optimal solution (solution is unique if there are distinct x_i 's)

$$\hat{\beta}_1 = \frac{s_{x,y}}{s_{x,x}} \quad \text{and} \quad \hat{\beta}_0 = \bar{y} - \bar{x}\hat{\beta}_1$$

$$\text{where } \bar{a} = \frac{1}{n} \sum_{i=1}^n a_i \text{ and } s_{a,b} = \sum_{i=1}^n (a_i - \bar{a})(b_i - \bar{b}).$$

- Fitted model
 - **Fitted values** $\hat{y}_i = \hat{\beta}_1 x_i + \hat{\beta}_0$, **residuals** $\hat{\varepsilon}_i = y_i - \hat{y}_i$.
 - **Error sum of squares (SSE)** and **total sum of squares (SST)**

$$\text{SSE} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad \text{and} \quad \text{SST} = \sum_{i=1}^n (y_i - \bar{y})^2$$

- * Interpretation of SST: variations of the response variable y ($\frac{1}{n-1} \text{SST} \approx \text{its variance}$);
- * Interpretation of SSE: the variations in y that the linear model does not explain.
 - One can further use SSE to estimate $\sigma^2 = \mathbb{E}[\varepsilon^2]$.
 - **Residual standard error** $\hat{\sigma} = \sqrt{\frac{\text{SSE}}{n-2}}$.

* These are absolute values.

- **coefficient of determination** / R^2

$$r^2 = 1 - \frac{\text{SSE}}{\text{SST}}$$

- * Interpretation of R^2 : how much proportion of y variations can be explained by linear model.
- * $R^2 = 0 \rightsquigarrow \hat{\beta}_1 = 0$, $R^2 = 1 \rightsquigarrow y_i = \hat{\beta}_1 x_i + \hat{\beta}_0$.
- Prediction and Interpretation
 - * Given some new input x_{test} , use $y_{\text{test}} = \hat{\beta}_1 x_{\text{test}} + \hat{\beta}_0$ to predict the response variable.

- * Interpretation of β_1 : One can observe the expectation of y goes up by β_1 if when x goes up by one unit in the observational data.
 - This is the association based on observed data, not causality.
 - For causality, we hope if we get involved, increasing one unit of x will lead to a β_1 increase in y .

2. **(Interpretation of linear model in the real world)** A quantitative researcher wants to get some preliminary understanding of two variables x and y using observations and linear least squares. He/She first tries to regress y on x using data in the whole day and gets a significant positive $\hat{\beta}_{1,full}$ (which means $\hat{\beta}_{1,full} > 0$ and SSE is significantly smaller than SST). Then he/she tries to regress y on x using data in the morning/noon/afternoon respectively and gets a significantly negative $\hat{\beta}_{1,m}$, $\hat{\beta}_{1,n}$ and $\hat{\beta}_{1,a}$.

[Q]: Is it possible that the above scenario will happen? If the answer is no, please provide formal proof. If the answer is yes, please explain the intuitions (for example, a geometric illustration or specify a concrete example) and which association (negative/positive) he/she can use in the real world.

3. **(Simple Linear Regression, Exercise)** The sales of a company (in million dollars) for each year are shown in the table below.

x (year)	2005	2006	2007	2008	2009
y (sales)	12	19	29	37	45

- Find the least square regression line $y = \hat{\beta}_0 + \hat{\beta}_1 x$.
- Use the least squares regression line as a model to estimate the sales of the company in 2012.